

U-Statistics

1.1 Definition

Definition 1.1. Suppose $(Y_i)_{i=1}^n$ is a random sample and $h : \mathbb{R}^m \rightarrow \mathbb{R}$ is a symmetric *kernel* function with:

$$\theta \equiv \mathbb{E}\{h(Y_1, \dots, Y_m)\}.$$

Here symmetric means $h(\cdot)$ is invariant with respect to permutations $(i_1, \dots, i_m)$ of the indices $(1, \dots, m)$. The **U-statistic** induced by h is:

$$U_n(h) = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} h(Y_{i_1}, \dots, Y_{i_m}). \quad (1.1)$$

■

Remark 1.1. Any unbiased estimator $u(Y_1, \dots, Y_m)$ can be symmetrized via:

$$h(Y_1, \dots, Y_m) \equiv \frac{1}{m!} \sum_P u(Y_{i_1}, \dots, Y_{i_m}),$$

where $\sum_P$ denotes the sum across all permutations of the indices $(i_1, \dots, i_m)$. ◆

Proposition 1.1. The definition of the U -statistic in (1.1) is equivalent to:

$$U_n = \mathbb{E}\{h(Y_1, \dots, Y_m) | (Y_{(1)}, \dots, Y_{(n)})\}.$$

That is, $U_n(h)$ may be regarded as the expectation of $h(Y_1, \dots, Y_m)$ conditional on the order statistics $(Y_{(1)}, \dots, Y_{(n)})$. ◆

Proof. Given the order statistics, what remains random when evaluating the expectation of $h(Y_1, \dots, Y_m)$ is the selection of the indices for the m arguments from among the n possible values. This can be done in n choose m possible ways. Let $(i_1, \dots, i_m)$ denote one such sample of indices. Since h is symmetric in its arguments, the sample of indices may always be arranged such that $1 \leq i_1 < \dots < i_m \leq n$. Therefore:

$$\mathbb{E}\{h(Y_1, \dots, Y_m) | Y_{(1)}, \dots, Y_{(n)}\} = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} h(Y_{i_1}, \dots, Y_{i_m}).$$

■

1.2 Examples

Example 1.1.

- The sample mean is a U -statistic of order $m = 1$:

$$U_n = \frac{1}{n} \sum_{1 \leq i \leq n} Y_i.$$

- The sample variance is a U -statistic of order $m = 2$:

$$U_n = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} \frac{1}{2} (Y_i - Y_j)^2.$$

- The Gini mean absolute difference is a U -statistic of order $m = 2$:

$$U_n = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} |Y_i - Y_j|.$$



1.3 Optimality

Theorem 1.1. Suppose $S(Y_1, \dots, Y_m)$ is unbiased for θ . The U -statistic induced by S , defined by $U_n = \mathbb{E}\{S(Y_1, \dots, Y_m) | (Y_{(1)}, \dots, Y_{(n)})\}$, is uniformly better than S in that sense that U_n is unbiased for θ and:

$$\mathbb{V}(U_n) \leq \mathbb{V}(S).$$

Moreover, the inequality is strict unless $U_n = S$. □

Proof. By iterated expectation, U_n is unbiased:

$$\mathbb{E}(U_n) = \mathbb{E}\mathbb{E}\{S(Y_1, \dots, Y_m) | (Y_{(1)}, \dots, Y_{(n)})\} = \mathbb{E}(S) = \theta.$$

By law of total variance:

$$\begin{aligned} \mathbb{V}(S) &= \text{Var}\left[E\{S | (Y_{(1)}, \dots, Y_{(n)})\}\right] + E\left[\text{Var}\{S | (Y_{(1)}, \dots, Y_{(n)})\}\right] \\ &= \mathbb{V}(U_n) + E\left[\text{Var}\{S | (Y_{(1)}, \dots, Y_{(n)})\}\right]. \end{aligned}$$

Since the latter term is non-negative, $\mathbb{V}(U_n) \leq \mathbb{V}(S)$. ■

1.4 Properties

Definition 1.2. Consider $U_n(h)$ from (1.1). For $1 \leq k \leq m$, define the **projection**:

$$h_k(y_1, \dots, y_k) = \mathbb{E}\{h(Y_1, \dots, Y_m) | Y_1 = y_1, \dots, Y_k = y_k\}.$$

Let ζ_k denote the variance of the k th projection:

$$\zeta_k = \mathbb{V}\{h_k(Y_1, \dots, Y_k)\}.$$

■

Remark 1.2. Note that the m th projection is:

$$h_m = \mathbb{E}\{h(Y_1, \dots, Y_m) | Y_1 = y_1, \dots, Y_m = y_m\} = h(y_1, \dots, y_m),$$

and that $\zeta_m = \mathbb{V}(h_m) = \mathbb{V}\{h(Y_1, \dots, Y_m)\}$.

◆

Proposition 1.2. Each projection h_k has the same expectation as $U_n(h)$:

$$\mathbb{E}\{h_k(Y_1, \dots, Y_k)\} = \theta.$$

◆

Proof.

$$\begin{aligned} \mathbb{E}\{h_k(Y_1, \dots, Y_k)\} &= E\left[\mathbb{E}\{h(Y_1, \dots, Y_m) | Y_1 = y_1, \dots, Y_k = y_k\}\right] \\ &= \mathbb{E}\{h(Y_1, \dots, Y_m)\} = \theta. \end{aligned}$$

■

Proposition 1.3. The variance of the projection h_k is increasing in k :

$$0 \leq \zeta_1 \leq \dots \leq \zeta_m = \mathbb{V}(U).$$

◆

Proof.

$$\begin{aligned} \mathbb{V}\{h_k(Y_1, \dots, Y_k)\} &= \mathbb{V}\left[\mathbb{E}\{h_k(Y_1, \dots, Y_k) | (Y_1, \dots, Y_{k-1})\}\right] \\ &\quad + \mathbb{E}\left[\mathbb{V}\{h_k(Y_1, \dots, Y_k) | (Y_1, \dots, Y_{k-1})\}\right] \\ &\geq \mathbb{V}\{h_{k-1}(Y_1, \dots, Y_{k-1})\} = \zeta_{k-1}. \end{aligned}$$

■

Proposition 1.4. Suppose $(Y_i)_{i=1}^n$ are a random sample, then:

$$\zeta_k = \mathbb{C}\{h(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m), \\ h(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m)\}.$$

Here $(Y_1, \dots, Y_k)$ are in common, while $(Y_{k+1}, \dots, Y_m)$ are distinct from $(\tilde{Y}_{k+1}, \dots, \tilde{Y}_m)$. ◆

Proof. By definition:

$$\begin{aligned} \gamma &\equiv \mathbb{C}\{h(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m), h(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m)\} \\ &= \mathbb{E}\{h(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m)h(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m)\} \\ &\quad - \mathbb{E}\{h(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m)\}\mathbb{E}\{h(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m)\}. \end{aligned}$$

The expectation of each kernel function is θ :

$$\gamma = \mathbb{E}\{h(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m)h(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m)\} - \theta^2.$$

By iterated expectation:

$$\begin{aligned} &E\{h(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m)h(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m)\} \\ &= \mathbb{E}\{h(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m)h(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m) | (Y_1, \dots, Y_k)\} \\ &= \mathbb{E}\{h_k(Y_1, \dots, Y_k)h_k(Y_1, \dots, Y_k)\}. \end{aligned}$$

Now:

$$\begin{aligned} \gamma &= \mathbb{E}\{h_k(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_m)h_k(Y_1, \dots, Y_k, \tilde{Y}_{k+1}, \dots, \tilde{Y}_m)\} - \theta^2 \\ &= \mathbb{E}\{h_k^2(Y_1, \dots, Y_k)\} - E^2\{h_k(Y_1, \dots, Y_k)\} = \zeta_k. \end{aligned}$$
■

Theorem 1.2 (Finite Sample Variance). The finite sample variance of the U -statistic in (1.1) is:

$$\mathbb{V}\{U_n(h)\} = \binom{n}{m}^{-1} \sum_{k=1}^m \binom{m}{k} \binom{n-m}{m-k} \zeta_k. \quad (1.2)$$

□

Proof.

$$\begin{aligned} \mathbb{V}\{U_n(h)\} &= \mathbb{V}\left\{\binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} h(Y_{i_1}, \dots, Y_{i_m})\right\} \\ &= \binom{n}{m}^{-2} \mathbb{C}\left\{\sum_{1 \leq i_1 < \dots < i_m \leq n} h(Y_{i_1}, \dots, Y_{i_m}), \sum_{1 \leq j_1 < \dots < j_m \leq n} h(Y_{j_1}, \dots, Y_{j_m})\right\} \\ &= \binom{n}{m}^{-2} \sum_{1 \leq i_1 < \dots < i_m \leq n} \sum_{1 \leq j_1 < \dots < j_m \leq n} \mathbb{C}\{h(Y_{i_1}, \dots, Y_{i_m}), h(Y_{j_1}, \dots, Y_{j_m})\} \end{aligned}$$

Consider cases. If $\{i_1, \dots, i_m\}$ and $\{j_1, \dots, j_m\}$ have no indices in common, then:

$$\mathbb{C}\{h(Y_{i_1}, \dots, Y_{i_m}), h(Y_{j_1}, \dots, Y_{j_m})\} = 0.$$

Suppose $\{i_1, \dots, i_m\}$ and $\{j_1, \dots, j_m\}$ have $1 \leq k \leq m$ indices in common, then:

$$\mathbb{C}\{h(Y_{i_1}, \dots, Y_{i_m}), h(Y_{j_1}, \dots, Y_{j_m})\} = \zeta_k$$

The number of covariance terms with k indices in common is:

$$\binom{n}{m} \binom{m}{k} \binom{n-m}{m-k}.$$

$\binom{n}{m}$ is the number of ways to choose the first index set $\{i_1, \dots, i_m\}$. Having fixed the first set, there are $\binom{m}{k}$ ways to select the k indices $\{j_1, \dots, j_k\}$ of the second set that are in common, times $\binom{n-m}{m-k}$ ways to select the $m-k$ indices $\{j_{k+1}, \dots, j_m\}$ of the second set that are different. Overall:

$$\mathbb{V}\{U_n(h)\} = \binom{n}{m}^{-2} \sum_{k=1}^m \binom{n}{m} \binom{m}{k} \binom{n-m}{m-k} \zeta_k.$$

■

Proposition 1.5 (Limiting Variance). If $\zeta_1 > 0$ and $\zeta_m < \infty$, then:

$$\lim_{n \rightarrow \infty} \mathbb{V}\{\sqrt{n}(U_n - \theta)\} = m^2 \zeta_1. \quad (1.3)$$

◆

Proof. From (1.2):

$$\mathbb{V}\{U_n(h)\} = \binom{n}{m}^{-1} \left\{ \binom{m}{1} \binom{n-m}{m-1} \zeta_1 + \binom{m}{2} \binom{n-m}{m-2} \zeta_2 + \dots \right\}.$$

For $n \rightarrow \infty$, the binomial coefficient is:

$$\binom{n-m}{a} = \frac{(n-m)!}{a!(n-m-a)!} = \frac{1}{a!} (n-m)(n-m-1) \cdots (n-m-a+1) \asymp \frac{n^a}{a!}$$

Thus, to leading order:

$$\mathbb{V}\{U_n(h)\} \asymp \frac{m!}{n^m} \left\{ m \cdot \frac{n^{m-1}}{(m-1)!} \zeta_1 + \mathcal{O}(n^{m-2}) \right\} = \frac{m^2 \zeta_1}{n} + \mathcal{O}(n^{-2}).$$

Conclude that: $\mathbb{V}\{\sqrt{n}(U_n - \theta)\} = m^2 \zeta_1 + \mathcal{O}(n^{-1})$.

■

Theorem 1.3 (Asymptotic Normality). Consider the U -statistic in (1.1).

i. If $0 < \zeta_1 < \infty$, then as $n \rightarrow \infty$:

$$\sqrt{n}(U_n - \theta) \xrightarrow{\mathcal{L}} N(0, \sigma^2),$$

where $\sigma^2 = m^2 \zeta_1$ is the limiting variance in (1.3).

ii. If $0 < \zeta_1 \leq \zeta_m < \infty$, then as $n \rightarrow \infty$:

$$\frac{U_n - \theta}{\sqrt{\mathbb{V}(U_n)}} \xrightarrow{\mathcal{L}} N(0, 1),$$

where $\mathbb{V}(U_n)$ is the finite sample variance in (1.2).

□

Proof. (ii.) Consider the Hajék projection representation of U_n from (2.2):

$$\hat{S} - \theta = \frac{m}{n} \sum_{i=1}^n \{h_1(Y_i) - \theta\}$$

The variance of $(\hat{S} - \theta)$ is:

$$\mathbb{V}(\hat{S} - \theta) = \frac{m^2}{n} \mathbb{V}\{h_1(Y_i)\} = \frac{m^2}{n} \zeta_1.$$

By the central limit theorem:

$$\frac{\hat{S} - \theta}{\sqrt{m^2 \zeta_1 / n}} \xrightarrow{\mathcal{L}} N(0, 1).$$

By (2.1), if $\lim_{n \rightarrow \infty} \mathbb{V}(U_n - \theta) / \mathbb{V}(\hat{S} - \theta) = 1$, then:

$$\frac{\hat{S} - \theta}{\sqrt{\mathbb{V}(\hat{S} - \theta)}} = \frac{U_n - \theta}{\sqrt{\mathbb{V}(U_n - \theta)}} + o_p(1).$$

From (1.3), the variance of $(U_n - \theta)$ is:

$$\mathbb{V}(U_n - \theta) = \frac{m^2}{n} \zeta_1 + \mathcal{O}(n^{-2}).$$

The limiting variances are asymptotically equivalent, since:

$$\frac{\frac{m^2}{n} \zeta_1 + \mathcal{O}(n^{-2})}{\frac{m^2}{n} \zeta_1} = 1 + \mathcal{O}(n^{-1}).$$

Adding and subtracting the standardized projection:

$$\frac{U_n - \theta}{\sqrt{\mathbb{V}(U_n - \theta)}} = \left(\frac{U_n - \theta}{\sqrt{\mathbb{V}(U_n - \theta)}} - \frac{\hat{S} - \theta}{\sqrt{\mathbb{V}(\hat{S} - \theta)}} \right) + \frac{\hat{S} - \theta}{\sqrt{\mathbb{V}(\hat{S} - \theta)}}$$

By (2.1), the term in parentheses converges in probability to zero, and by the central limit theorem, the standardized projection converges in distribution to standard normal. From Slutsky's theorem, conclude that:

$$\frac{U_n - \theta}{\sqrt{\mathbb{V}(U_n - \theta)}} \xrightarrow{\mathcal{L}} N(0, 1).$$

■

1.5 Wilcoxon One-Sample Statistic

Example 1.2. Suppose $(Y_i) \stackrel{\text{iid}}{\sim} F_Y$. Consider the parameter $\theta = P(Y_1 + Y_2 > 0)$. The corresponding U -statistic is:

$$U_n = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} \mathbb{I}(Y_i + Y_j > 0). \quad (1.4)$$

Identify $h(Y_1, Y_2) = \mathbb{I}(Y_1 + Y_2 > 0)$ as the kernel. The first projection of h is:

$$\begin{aligned} h_1(y_1) &= \mathbb{E}\{h(Y_1, Y_2) | Y_1 = y_1\} = \mathbb{E}\{\mathbb{I}(y_1 + Y_2 > 0)\} \\ &= P(Y_2 > -y_1) = 1 - F_Y(-y_1). \end{aligned}$$

The Hajék representation of U_n is:

$$\hat{S} - \theta = \frac{2}{n} \sum_{i=1}^n \{1 - F_Y(-Y_i) - \theta\},$$

where $\theta = \mathbb{E}\{1 - F_Y(-Y)\}$. Suppose that, under H_0 , F_Y is *continuous* and *symmetric*, then by the probability integral theorem $F_Y(Y) \sim U(0, 1)$, and:

$$1 - F_Y(-Y) = F_Y(Y) \stackrel{d}{=} U(0, 1).$$

Now, under H_0 ,

$$\begin{aligned} \theta &= \mathbb{E}\{1 - F_Y(-Y)\} = \mathbb{E}\{U(0, 1)\} = \frac{1}{2}, \\ \zeta_1 &= \mathbb{V}\{h_1(Y)\} = \mathbb{V}\{U(0, 1)\} = \frac{1}{12}. \end{aligned}$$

Therefore, under H_0 ,

$$\sqrt{n} \left(U_n - \frac{1}{2} \right) \xrightarrow{\mathcal{L}} N \left(0, \frac{4}{12} \right) = N \left(0, \frac{1}{3} \right)$$

The Wilcoxon signed-rank statistic is:

$$W_n = \sum_{i=1}^n \text{rank}(|Y_i|) \mathbb{I}(Y_i > 0), \quad \text{rank}(|Y_i|) = \sum_{j=1}^n \mathbb{I}(|Y_j| \leq |Y_i|).$$

Under the null hypothesis that F_Y is *continuous*, such that there are no ties, the normalized Wilcoxon one-sample statistic $\binom{n}{2}^{-1} W_n$ is asymptotically equivalent to the U -statistic in (1.4). To see this, first rewrite W_n using the indicator definition of rank:

$$W_n = \sum_{i=1}^n \sum_{j=1}^n \mathbb{I}(|Y_j| \leq |Y_i|) \mathbb{I}(Y_i > 0).$$

The summand evaluates to 1 if and only if $|Y_j| \leq Y_i$:

$$W_n = \sum_{i=1}^n \sum_{j=1}^n \mathbb{I}(|Y_j| \leq Y_i).$$

Split the domain of summation:

$$W_n = \sum_{i < j} \mathbb{I}(|X_j| < X_i) + \sum_{i < j} \mathbb{I}(|X_i| < X_j) + \sum_{i=1}^n \mathbb{I}(Y_i > 0).$$

The last term arises because $|Y_j| \leq Y_i$ only holds for positive Y_i . Let $P_n = \sum_{i=1}^n \mathbb{I}(Y_i > 0)$ and combine the two leading sums:

$$\begin{aligned} W_n &= \sum_{i < j} \{ \mathbb{I}(|X_j| < X_i) + \mathbb{I}(|X_i| < X_j) \} + P_n \\ &= \sum_{i < j} \{ \mathbb{I}(X_i - |X_j| > 0) + \mathbb{I}(X_j - |X_i| > 0) \} + P_n. \end{aligned}$$

Observe that the term in braces $\{ \}$ evaluates to 1 if and only if $X_i + X_j > 0$, therefore:

$$W_n = \sum_{1 \leq i < j \leq n} \mathbb{I}(X_i + X_j > 0) + P_n.$$

Note that P_n is bounded by n . Upon normalizing by n choose 2:

$$\binom{n}{2}^{-1} W_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \mathbb{I}(X_i + X_j > 0) + \binom{n}{2}^{-1} P_n = U_n + \mathcal{O}_p(n^{-1}).$$



1.6 Multi-sample Extension

Definition 1.3. Suppose $(X_i)_{i=1}^{n_1}$ and $(Y_i)_{i=1}^{n_2}$ are random samples, and $h : \mathbb{R}^m \rightarrow \mathbb{R}$ is a kernel function ($m = m_1 + m_2$), symmetric within groups of arguments, and:

$$\theta \equiv \mathbb{E}\{h(X_1, \dots, X_{m_1}; Y_1, \dots, Y_{m_2})\}.$$

The two-sample **U-statistic** ($n = n_1 + n_2$) takes the form:

$$U_n(h) = \frac{1}{\binom{n_1}{m_1} \binom{n_2}{m_2}} \sum_{1 \leq i_1 < \dots < i_{m_1} \leq n_1} \sum_{1 \leq j_1 < \dots < j_{m_2} \leq n_2} h(X_{i_1}, \dots, X_{i_{m_1}}; Y_{j_1}, \dots, Y_{j_{m_2}}). \quad (1.5)$$

Extensions to addition samples are analogous. ■

Definition 1.4. Consider the multi-sample U -statistic in (1.5). Define ζ_{kl} as:

$$\zeta_{kl} = \mathbb{C}\{h(X_1, \dots, X_k, X_{k+1}, \dots, X_{m_1}; Y_1, \dots, Y_l, Y_{l+1}, \dots, Y_{m_2}), \\ h(X_1, \dots, X_k, \tilde{X}_{k+1}, \dots, \tilde{X}_{m_1}; Y_1, \dots, Y_l, \tilde{Y}_{l+1}, \dots, \tilde{Y}_{m_2})\}$$

Here $(X_1, \dots, X_k)$ and $(Y_1, \dots, Y_l)$ are in common, while $(X_{k+1}, \dots, X_{m_1})$ are distinct from $(\tilde{X}_{k+1}, \dots, \tilde{X}_{m_1})$, and $(Y_{l+1}, \dots, Y_{m_2})$ are distinct from $(\tilde{Y}_{l+1}, \dots, \tilde{Y}_{m_2})$. ■

Theorem 1.4 (Finite Sample Variance). The finite sample variance of the U -statistic in (1.5) is:

$$\mathbb{V}\{U_n(h)\} = \sum_{k=0}^{m_1} \sum_{l=0}^{m_2} \frac{\binom{m_1}{k} \binom{n_1-m_1}{m_1-k}}{\binom{n_1}{m_1}} \frac{\binom{m_2}{l} \binom{n_2-m_2}{m_2-l}}{\binom{n_2}{m_2}} \zeta_{kl}. \quad (1.6)$$

□

Theorem 1.5 (Limiting Variance). Consider the U -statistic in (1.5). Suppose $(n_1, n_2) \rightarrow \infty$ in such a way that:

$$\lim_{(n_1, n_2) \rightarrow \infty} \frac{n_1}{n_1 + n_2} = \rho, \quad \lim_{(n_1, n_2) \rightarrow \infty} \frac{n_2}{n_1 + n_2} = 1 - \rho,$$

and let $n = n_1 + n_2$. If $0 < \zeta_{10}, \zeta_{01} \leq \zeta_{m_1 m_2} < \infty$, then:

$$\sigma^2 = \lim_{n \rightarrow \infty} \mathbb{V}\{\sqrt{n}(U_n - \theta)\} = \frac{m_1^2}{\rho} \zeta_{10} + \frac{m_2^2}{1 - \rho} \zeta_{01}. \quad (1.7)$$

□

Remark 1.3. See Lehmann (1999), theorem 6.1.3. ◆

Theorem 1.6 (Asymptotic Normality). Consider the multi-sample U -statistic in (1.5).

i. If $0 < \zeta_{10}, \zeta_{01} < \infty$, then as $n \rightarrow \infty$:

$$\sqrt{n}(\hat{U}_n - \theta) \xrightarrow{\mathcal{L}} N(0, \sigma^2),$$

where σ^2 is the limiting variance in (1.7).

ii. If $0 < \zeta_{10}, \zeta_{01} \leq \zeta_{m_1 m_2} < \infty$, then:

$$\frac{\hat{U}_n - \theta}{\sqrt{\mathbb{V}(U_n)}} \xrightarrow{\mathcal{L}} N(0, 1),$$

where $\mathbb{V}(U_n)$ is the finite sample variance in (1.6).

□

Remark 1.4. See Lehmann (1999), theorem 6.1.4.

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1.7 Mann-Whitney Statistic

Example 1.3. Suppose $(X_i) \stackrel{\text{iid}}{\sim} F_X$ and $(Y_j) \stackrel{\text{iid}}{\sim} F_Y$. Consider the parameter $\theta = P(X \leq Y)$. The corresponding U -statistic is:

$$U_n = \frac{1}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbb{I}(X_i \leq Y_j).$$

Identify $h(X, Y) = \mathbb{I}(X \leq Y)$ as the kernel. The first projections of h are:

$$h_{10}(x) = \mathbb{E}\{h(X, Y)|X = x\} = P(x \leq Y) = 1 - F_Y(x)$$

$$h_{01}(y) = \mathbb{E}\{h(X, Y)|Y = y\} = P(X \leq y) = F_X(y).$$

Suppose that, under H_0 , $F_X = F_Y = F_0$, with F_0 continuous, then by the probability integral theorem $h_{10}(X) = 1 - F_0(X) \sim U(0, 1)$ and $h_{01}(Y) = F_0(Y) \sim U(0, 1)$.

Now, under H_0 ,

$$\theta = \mathbb{E}\{\mathbb{I}(X \leq Y)\} = \mathbb{E}\{F_0(Y)\} = \mathbb{E}\{U(0, 1)\} = \frac{1}{2},$$

$$\zeta_{10} = \mathbb{V}\{h_{10}(X)\} = \mathbb{V}\{U(0, 1)\} = \frac{1}{12},$$

$$\zeta_{01} = \mathbb{V}\{h_{01}(Y)\} = \mathbb{V}\{U(0, 1)\} = \frac{1}{12}.$$

Therefore, under H_0 ,

$$\sqrt{n} \left(U_n - \frac{1}{2} \right) \xrightarrow{\mathcal{L}} N \left\{ 0, \frac{1}{12} \left(\frac{1}{\rho} + \frac{1}{1-\rho} \right) \right\},$$

where $n = n_1 + n_2$, and $\rho = \lim_{n \rightarrow \infty} n_1/n$.

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1.8 Joint Distribution

Theorem 1.7 (Joint Asymptotic Normality). Suppose $(Y_i)_{i=1}^n$ is a random sample, and consider two distinct, one-sample U -statistics $(U_n^{(1)}, U_n^{(2)})$ of the form (1.1). Let $\theta_k = E(U_n^{(k)})$, and define:

$$\gamma_k = \mathbb{C}\{h^{(1)}(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_{m_1}), \\ h^{(2)}(Y_1, \dots, Y_k, Y_{k+1}, \dots, Y_{m_2})\},$$

where $h^{(k)}$ is the *kernel* of $U_n^{(k)}$. If

$$0 < \mathbb{V}\{h^{(1)}(Y_1, \dots, Y_{m_1})\} < \infty, \quad 0 < \mathbb{V}\{h^{(2)}(Y_1, \dots, Y_{m_2})\} < \infty,$$

then:

$$\sqrt{n} \begin{pmatrix} U_n^{(1)} - \theta_1 \\ U_n^{(2)} - \theta_2 \end{pmatrix} \xrightarrow{\mathcal{L}} N \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} m_1^2 \zeta_1^{(1)} & m_1 m_2 \gamma_1 \\ m_1 m_2 \gamma_1 & m_2^2 \zeta_1^{(2)} \end{pmatrix} \right\}$$

where $\zeta_1^{(1)} = \mathbb{V}\{h_1^{(1)}(Y_1)\}$ and $\zeta_2^{(2)} = \mathbb{V}\{h_1^{(2)}(Y_1)\}$. □

Remark 1.5. See Lehmann (1999), theorem 6.1.6. ◆

Example 1.4. Suppose $(Y_i) \stackrel{\text{iid}}{\sim} F_Y$. Consider the joint asymptotic distribution of the sample mean and variance:

$$U_n^{(1)} = \frac{1}{n} \sum_{i=1}^n Y_i, \quad U_n^{(2)} = \frac{2}{n(n-1)} \sum_{i < j} \frac{1}{2} (Y_i - Y_j)^2.$$

For the mean sample mean $h_1^{(1)}(Y_1) = h^{(1)}(Y_1) = Y_1$, hence:

$$\theta_1 = \mathbb{E}\{h_1^{(1)}(Y_1)\} = \mathbb{E}\{Y_1\} = \mu, \\ \zeta_1^{(1)} = \mathbb{V}\{h_1^{(1)}(Y_1)\} = \mathbb{V}(Y_1) = \sigma^2.$$

For the variance, the first projection is:

$$h_1^{(2)}(y) = \mathbb{E}\{h^{(2)}(Y_1, Y_2) | Y_1 = y\} = \frac{1}{2} \mathbb{E}\{(y - Y_2)^2\} = \frac{1}{2} \mathbb{E}\{(Y_2 - \mu + \mu - y)^2\} \\ = \frac{1}{2} \mathbb{E}\{(Y_2 - \mu)^2\} + \frac{1}{2} (y - \mu)^2 = \frac{1}{2} \sigma^2 + \frac{1}{2} (y - \mu)^2.$$

Thus:

$$\theta_2 = \mathbb{E}\{h_1^{(2)}(Y_1)\} = \frac{1}{2} \sigma^2 + \frac{1}{2} \mathbb{E}\{(Y_1 - \mu)^2\} = \sigma^2, \\ \zeta_1^{(2)} = \mathbb{V}\{h_1^{(2)}(Y_1)\} = \frac{1}{4} \mathbb{V}\{(Y_1 - \mu)^2\} = \frac{1}{4} (\mu_4 - \sigma^4),$$

where $\mu_4 = \mathbb{E}\{(Y_1 - \mu)^4\}$.

The covariance term is:

$$\begin{aligned}\gamma &= \mathbb{C}\{h^{(1)}(Y_1), h^{(2)}(Y_1, Y_2)\} = \frac{1}{2}\mathbb{C}\{Y_1, (Y_1 - Y_2)^2\} \\ &= \frac{1}{2}\mathbb{C}\{(Y_1 - \mu), (Y_1 - \mu + \mu - Y_2)^2\} \\ &= \frac{1}{2}\mathbb{C}\{(Y_1 - \mu), (Y_1 - \mu)^2 + (Y_2 - \mu)^2 - 2(Y_1 - \mu)(Y_2 - \mu)\}\end{aligned}$$

By independence of Y_1 from Y_2 :

$$\gamma = \frac{1}{2}\mathbb{C}\{(Y_1 - \mu), (Y_1 - \mu)^2\} - \mathbb{C}\{(Y_1 - \mu), (Y_1 - \mu)(Y_2 - \mu)\} = \frac{1}{2}\mu_3 - 0,$$

where $\mu_3 = \mathbb{E}\{(Y_1 - \mu)^3\}$.

Overall, noting that $m_1 = 1$ and $m_2 = 2$, the joint asymptotic distribution of the sample mean and covariance is:

$$\sqrt{n} \begin{pmatrix} U_n^{(1)} - \mu \\ U_n^{(2)} - \sigma^2 \end{pmatrix} \xrightarrow{\mathcal{L}} N \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \mu_3 \\ \mu_3 & \mu_4 - \sigma^4 \end{pmatrix} \right\}.$$

This result may be confirmed using the M -estimation framework. ♠

Projection

2.1 Projections of Random Variables

Definition 2.1. Let $\mathcal{V}$ denote the linear space of random variables with finite second moments, and let $\mathcal{U}$ denote a closed linear subspace. The projection of $V \in \mathcal{V}$ onto $\mathcal{U}$ is the random variable $\hat{U} = \Pi(V|\mathcal{U})$ satisfying:

- i. $\hat{U} \in \mathcal{U}$,
 - ii. $\mathbb{E}\{(V - \hat{U})U\} = 0$ for $\forall U \in \mathcal{U}$.
-

Proposition 2.1. Suppose $V \in \mathcal{V}$, then $\hat{U} = \Pi(V|\mathcal{U})$ is the closest element in $\mathcal{U}$ to V in the sense that:

$$\mathbb{E}(\hat{U} - V)^2 \leq \mathbb{E}(U - V)^2 \text{ for } \forall U \in \mathcal{U}.$$

Moreover, the inequality is strict unless $\mathbb{E}(U - \hat{U})^2 = 0$. ♦

Proof.

$$\mathbb{E}(U - V)^2 = \mathbb{E}(U - \hat{U} + \hat{U} - V)^2 = \mathbb{E}(U - \hat{U})^2 + \mathbb{E}(\hat{U} - V)^2 + 2\mathbb{E}\{(U - \hat{U})(\hat{U} - V)\}.$$

Since $U \in \mathcal{U}$ and $\hat{U} \in \mathcal{U}$, the difference $(U - \hat{U}) \in \mathcal{U}^\perp$, thus:

$$\mathbb{E}\{(U - \hat{U})(\hat{U} - V)\} = 0.$$

Thus, $\mathbb{E}(U - V)^2 \geq \mathbb{E}(\hat{U} - V)^2$, equality holding if and only if $\mathbb{E}(U - \hat{U})^2 = 0$. ■

Proposition 2.2. Suppose the closed, linear subspace $\mathcal{U}$ contains the constant random variable, then:

- i. $\mathbb{E}(V) = \mathbb{E}(\hat{U})$,
- ii. $\mathbb{C}(V - \hat{U}, U) = 0$ for $\forall U \in \mathcal{U}$.
- iii. $\mathbb{C}(V, \hat{U}) = \mathbb{V}(\hat{U})$.

◆

Proof. (i.) If $\mathcal{U}$ contains the constant random variable $U = 1$, then the orthogonality condition implies $\mathbb{E}\{(V - \hat{U})1\} = \mathbb{E}(V - \hat{U}) = 0$, so $\mathbb{E}(V) = \mathbb{E}(\hat{U})$.

(ii.)

$$\mathbb{C}(V - \hat{U}, U) = \mathbb{E}\{(V - \hat{U})U\} - \mathbb{E}(V - \hat{U})\mathbb{E}(U) = 0.$$

(iii.) Since $(V - \hat{U})$ is orthogonal to $\hat{U}$:

$$\mathbb{E}\{(V - \hat{U})\hat{U}\} = 0 \implies \mathbb{E}(V\hat{U}) = \mathbb{E}(\hat{U}^2).$$

The covariance of V and $\hat{U}$ is:

$$\mathbb{C}(V, \hat{U}) = \mathbb{E}(V\hat{U}) - \mathbb{E}(V)\mathbb{E}(\hat{U}) = \mathbb{E}(\hat{U}^2) - \mathbb{E}^2(\hat{U}) = \mathbb{V}(\hat{U}).$$

■

Proposition 2.3. Suppose $\mathcal{U}_n$ is a sequence of closed linear subspaces, each containing the constant random variable. Let V_n denote a sequence of random variables with $\hat{U}_n = \Pi(V_n|\mathcal{U})$. If $\lim_{n \rightarrow \infty} \mathbb{V}(\hat{U}_n)/\mathbb{V}(V_n) = 1$, then:

$$\frac{V_n - \mathbb{E}(V_n)}{\sqrt{\mathbb{V}(V_n)}} = \frac{\hat{U}_n - \mathbb{E}(\hat{U}_n)}{\sqrt{\mathbb{V}(\hat{U}_n)}} + o_p(1). \quad (2.1)$$

◆

Proof. Recall that convergence in mean square implies convergence in probability. The variance of the difference is:

$$\mathbb{V} \left\{ \frac{V_n - \mathbb{E}(V_n)}{\sqrt{\mathbb{V}(V_n)}} - \frac{\hat{U}_n - \mathbb{E}(\hat{U}_n)}{\sqrt{\mathbb{V}(\hat{U}_n)}} \right\} = 1 + 1 - 2 \frac{\mathbb{C}(V_n, \hat{U}_n)}{\sqrt{\mathbb{V}(V_n)\mathbb{V}(\hat{U}_n)}}$$

Since $\mathbb{C}(V_n, \hat{U}_n) = \mathbb{V}(\hat{U}_n)$:

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{V} \left\{ \frac{V_n - \mathbb{E}(V_n)}{\sqrt{\mathbb{V}(V_n)}} - \frac{\hat{U}_n - \mathbb{E}(\hat{U}_n)}{\sqrt{\mathbb{V}(\hat{U}_n)}} \right\} &= 2 - 2 \lim_{n \rightarrow \infty} \frac{\mathbb{V}(\hat{U}_n)}{\sqrt{\mathbb{V}(V_n)\mathbb{V}(\hat{U}_n)}} \\ &= 2 - 2 \sqrt{\lim_{n \rightarrow \infty} \frac{\mathbb{V}(\hat{U}_n)}{\mathbb{V}(V_n)}} = 2 - 2 = 0. \end{aligned}$$

Thus the difference:

$$\frac{V_n - \mathbb{E}(V_n)}{\sqrt{\mathbb{V}(V_n)}} - \frac{\hat{U}_n - \mathbb{E}(\hat{U}_n)}{\sqrt{\mathbb{V}(\hat{U}_n)}} \xrightarrow{L^2} 0.$$

■

2.2 Projection onto Sums

Theorem 2.1. Suppose (Y_i) are independent. Consider the class $\mathcal{S}$ of random variables of the form:

$$S = \sum_{i=1}^n g_i(Y_i),$$

for measurable functions $g_i(\cdot)$ with finite second moments $\mathbb{E}\{g_i^2(Y_i)\}$. The **Hájek projection** of $V \in \mathcal{V}$ onto this class is given by:

$$\hat{S} = \Pi(V|\mathcal{S}) = \sum_{i=1}^n \{\mathbb{E}(V|Y_i)\} - (n-1)\mathbb{E}(V).$$

□

Proof. Towards finding $\mathbb{E}(\hat{S}|Y_j)$, consider the expectation of the random variable $\mathbb{E}(V|Y_i)$ given Y_j . For $i = j$:

$$\mathbb{E}\{\mathbb{E}(V|Y_i)|Y_j\} = \mathbb{E}(V|Y_i).$$

By independence, for $i \neq j$:

$$\mathbb{E}\{\mathbb{E}(V|Y_i)|Y_j\} = \mathbb{E}\{\mathbb{E}(V|Y_i)\} = \mathbb{E}(V).$$

Thus the expectation of the Hajek projection $\hat{S}$ given Y_j is:

$$\mathbb{E}(\hat{S}|Y_j) = \mathbb{E}(V|Y_j) + \sum_{i \neq j} \mathbb{E}\{\mathbb{E}(V|Y_i)|Y_j\} - (n-1)\mathbb{E}(V) = \mathbb{E}(V|Y_j).$$

For each j , the residual $(V - \hat{S})$ is orthogonal to $g_j(Y_j)$:

$$\begin{aligned} \mathbb{E}\{(V - \hat{S})g_j(Y_j)\} &= E\left[\mathbb{E}\{(V - \hat{S})g_j(Y_j)|Y_j\}\right] \\ &= \mathbb{E}\left[\{\mathbb{E}(V|Y_j) - \mathbb{E}(\hat{S}|Y_j)\}g_j(Y_j)\right] = 0. \end{aligned}$$

Thus $(V - \hat{S})$ is orthogonal to random variables of the form $S = \sum_{i=1}^n g_i(Y_i)$, i.e. to $\mathcal{S}$. Finally, since $\hat{S} \in \mathcal{S}$ and $(V - \hat{S})$ is orthogonal to $\mathcal{S}$, conclude that $\hat{S}$ is the projection of V onto $\mathcal{S}$. ■

Proposition 2.4. The Hajék projection of the U -statistic in (1.1) is:

$$\hat{S} - \theta = \frac{m}{n} \sum_{i=1}^n \{h_1(Y_i) - \theta\}. \quad (2.2)$$

◆

Proof. Recall that the U -statistic takes the form:

$$U_n(h) = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} h(Y_{i_1}, \dots, Y_{i_m}).$$

Taking the expectation conditional on Y_i :

$$E(U_n|Y_i = y) = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} \mathbb{E}\{h(Y_{i_1}, \dots, Y_{i_m})|Y_i = y\}$$

Consider cases. If $i \in \{i_1, \dots, i_m\}$, then:

$$\mathbb{E}\{h(Y_{i_1}, \dots, Y_{i_m})|Y_i = y\} = h_1(y),$$

whereas if $i \notin \{i_1, \dots, i_m\}$, then:

$$\mathbb{E}\{h(Y_{i_1}, \dots, Y_{i_m})|Y_i = y\} = \theta.$$

There are $\binom{n-1}{m-1}$ ways to choose $(i_1, \dots, i_m)$ so as to include i , and $\binom{n-1}{m}$ to choose $(i_1, \dots, i_m)$ so as to exclude i . Thus:

$$E(U_n|Y_i = y) = \binom{n}{m}^{-1} \left\{ \binom{n-1}{m-1} h_1(y) + \binom{n-1}{m} \theta \right\} = \frac{m}{n} h_1(y) + \frac{n-m}{n} \theta.$$

The Hajek projection is:

$$\hat{S} = \sum_{i=1}^n \left\{ \frac{m}{n} h_1(y) + \frac{n-m}{n} \theta \right\} - (n-1)\theta = \frac{m}{n} \sum_{i=1}^n \{h_1(Y_i) - \theta\} + n\theta - (n-1)\theta.$$

■

References

- Serfling. *Approximation Theorems of Mathematical Statistics* (1980); Chapter 5.
- van der Vaart. *Asymptotic Statistics* (1998); Chapters 11 & 12.
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